

Predicting Cardiotoxicity of Molecules Using Attention-Based Graph Neural Networks

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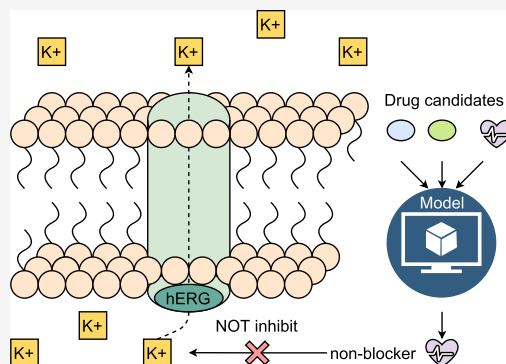


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ABSTRACT: In drug discovery, the search for new and effective medications is often hindered by concerns about toxicity. Numerous promising molecules fail to pass the later phases of drug development due to strict toxicity assessments. This challenge significantly increases the cost, time, and human effort needed to discover new therapeutic molecules. Additionally, a considerable number of drugs already on the market have been withdrawn or re-evaluated because of their unwanted side effects. Among the various types of toxicity, drug-induced heart damage is a severe adverse effect commonly associated with several medications, especially those used in cancer treatments. Although a number of computational approaches have been proposed to identify the cardiotoxicity of molecules, the performance and interpretability of the existing approaches are limited. In our study, we proposed a more effective computational framework to predict the cardiotoxicity of molecules using an attention-based graph neural network. Experimental results indicated that the proposed framework outperformed the other methods. The stability of the model was also confirmed by our experiments. To assist researchers in evaluating the cardiotoxicity of molecules, we have developed an easy-to-use online web server that incorporates our model.



1. INTRODUCTION

The ether-à-go-go-related gene (hERG) is essential for regulating the exchange of resting potentials and action potentials on the cardiac myocyte since it enables encryption of the pore-forming α -subunit of rapid delayed rectifier current.^{1,2} Blockage of the hERG channel probably leads to long-QT syndrome and even lethal cardiotoxicities, such as Torsades de Pointes arrhythmia, as confirmed by experimental and clinical evidence.³ Numerous medicines (e.g., ziprasidone, vardenafil, cisapride, terfenadine, and astemizole) have been withdrawn from the market or strictly controlled due to severe hERG-related cardiac adverse effects.^{4–6} In drug discovery, an early assessment of hERG blockage plays a crucial role since the hERG channel is incredibly sensitive to a large number of structurally diverse molecules as hERG blockers.^{7,8}

The International Conference on Harmonization issued a guideline to preclinically assess the hERG-inhibitory activities of drug candidates before submitting them for regulatory review.⁹ *In vitro* and *in vivo* screening techniques, such as radioligand binding, electrophysiology measurements, rubidium-flux assays, and fluorescence-based assays, have been prevalently used to investigate the hERG-blocking activities of drug candidates. These experimental approaches, however, require significant effort, time, and budget.¹⁰ Based on experimentally verified data, multiple *in silico* methods were developed as fast and cost-effective approaches that can be employed for computer-aided high-throughput screening in the early stages of drug discovery in terms of drug pharmacokinetics and pharmacodynamics.^{11–16}

In the past few years, a variety of prediction models have been developed to identify hERG blockers using various machine learning (ML) algorithms.^{5,6,17–20} Doddareddy et al. constructed classification models using linear discriminant analysis and support vector machines (SVMs) to predict hERG-related cardiotoxicity.²¹ Despite high performance on both training and test sets, Doddareddy et al.'s model has limited applicability due to the small size of the data set used. Wang et al. proposed classification models to detect hERG-active drugs using pharmacophore modeling in combination with ML approaches.²² In addition, many models were formed based on artificial decoy sets or molecules randomly selected from large chemical databases. The decoy set comprises completely inactive molecules designed under preset physiochemical constraints. Models trained by hERG blockers and decoys may not exhaustively learn specific features of hERG blockers, since decoys are far different from hERG blockers. On the other hand, the lack of experimental evidence for negative samples may cause modeling strategies to face higher risks of false positives. Besides, more advanced prediction models,²³ such as

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pkCSM,²⁴ DeepHIT,²⁵ CardioTox net,²⁶ and cardioToxCSSM,²⁷ were introduced to address this issue with promising outcomes. While pkCSM and cardioToxCSSM were developed using graph-based features,^{24,27} DeepHIT and CardioTox net were combinatorial models consisting of three separate neural networks trained with three types of features: molecular descriptors, fingerprints, and graph-based features.^{25,26} Although using graph-based features for modeling may not capture distinct substructural characteristics of compounds, these lightweight models can serve as high-throughput screening tools. Combinatorial or ensemble models can extract substructural characteristics to improve prediction efficiency; however, low-speed processing restricts their application in large-scale screening.

In our study, we propose a computational model to predict the cardiotoxicity of molecules using an attention-based graph neural network. Recently, graph neural networks have been widely employed for property prediction tasks with excellent results while accelerating the processing speed. Besides, the application of the attention mechanism has been demonstrated to significantly enhance prediction efficiency in numerous deep learning (DL) approaches. Therefore, the combination of the graph neural network and the attention mechanism is expected to create a robust, lightweight model with a competitive performance for large-scale screening. Our model is designed to identify six of the most common cardiotoxicities of molecules, including cardiac failure,²⁸ arrhythmia,²⁹ heart block,³⁰ myocardial infarction,³¹ hypertension,³² and hERG toxicity.³³ Our model is also deployed as an online web server to support the research community.

2. MATERIALS AND METHODS

2.1. Data Collection and Curation. The data sets used in this study were collected from Ifthakhar et al.²⁷ According to their description, a set of 18,803 molecules categorized into six types of cardiotoxicity—cardiac failure, arrhythmia, heart block, myocardial infarction, hypertension, and hERG toxicity—were collected from published studies, including Cai et al.,¹⁶ Munawar et al.,³⁴ and Karim et al.²⁶ The data samples in each data set were preprocessed by removing heavy-weight molecules (greater than 1 kDa) since oral drugs usually have smaller weights.³⁵ To curate these data sets, we followed the curation pipeline originally developed by Fourches et al.,³⁶ with some adjustments described in our previous work.³⁷ The pipeline comprises four principal phases: (i) validation, (ii) cleaning, (iii) normalization, and (iv) final verification (Figure 1).

During the initial phase, molecules identified as inorganic, mixture, or organometallic are excluded. The second phase involves the elimination of salts and the resolution of two types of charged molecules: those containing metals and organic ones. Metal-containing charged molecules are dismissed due to the presence of positively charged metal atoms, while organic charged molecules remain unaltered, as it is impractical to precisely delineate the experimental conditions under which these molecules exhibit activity. Consequently, organically charged molecules are neutralized. In the third phase, all tautomers and stereoisomers of identical molecules are converted into their distinct standardized forms to preclude the existence of interconvertible isomers known as “*detautomerization*” and “*destereoisomerization*”, respectively. Duplicates identified during these three initial phases are eliminated at the end of each respective phase. The fourth phase involves the removal of any chemical data (molecules) possessing conflicting

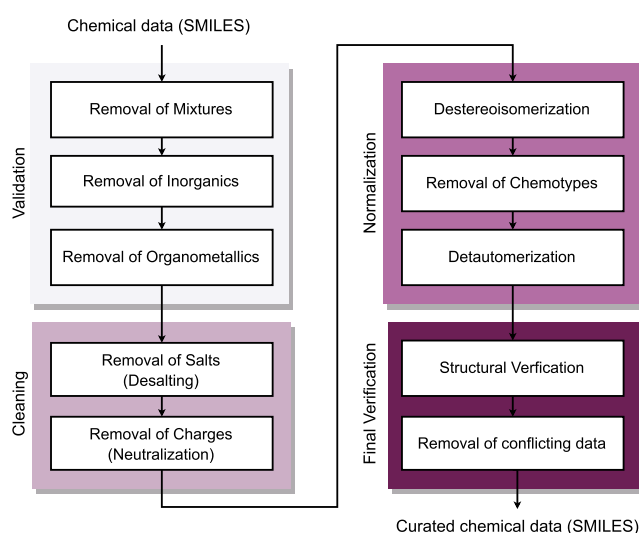


Figure 1. Overview of the data curation.

labels, or briefly designated as “*conflicting records*”. All molecular data were presented in canonical simplified molecular-input line-entry system (SMILES) format prior to entering any processing phases. Conflicting records can manifest in one of three cases: (a) multiple identical molecules with differing labels, (b) duplicate molecules with identical labels, or (c) multiple distinct molecules sharing the same CAS registry numbers. Records corresponding to cases (a) or (c) were discarded, while those matching case (b) were consolidated into a single entry. Additionally, all molecules were cross-verified using PubChem and ChEMBL databases, which are considered the most comprehensive and trustworthy chemical databases, to ensure the accuracy and validity of their structural identities.

After being curated, for each data set, the samples (molecules) were divided into a training set and a test set using stratified random sampling. The training set was then randomly split into five stratified folds in which each fold iteratively served as a validation set. The test sets were strictly controlled to not be involved in any stage of model training and validation. Principal component analysis was used to show the chemical space on the RDKit Molecular Descriptor (RDKitMD) features of the molecules in the training set, the test set, and between the classes for each data set (Figures S1–S6 in the Supporting Information). The visualizations confirm that the positives and negatives were chosen and assigned to the training and test sets fairly and representatively in the chemical space. Table 1 shows a summary of all the curated data sets.

2.2. Molecular Encoding Scheme. Figure 2 depicts the molecular encoding scheme employed in the present study. Each molecule, constituted by N heavy atoms (encompassing all potential atoms with the exclusion of hydrogen), M

Table 1. Datasets for Model Development and Evaluation

data set	number of samples (positives/negatives)		
	training	test	total
cardiac failure	264/722	30/80	294/802
arrhythmia	601/745	67/83	668/828
heart block	391/403	44/45	435/448
myocardial infarction	348/347	39/39	387/386
hypertension	595/584	67/65	662/649
hERG blocker	4822/4327	536/481	5358/4808

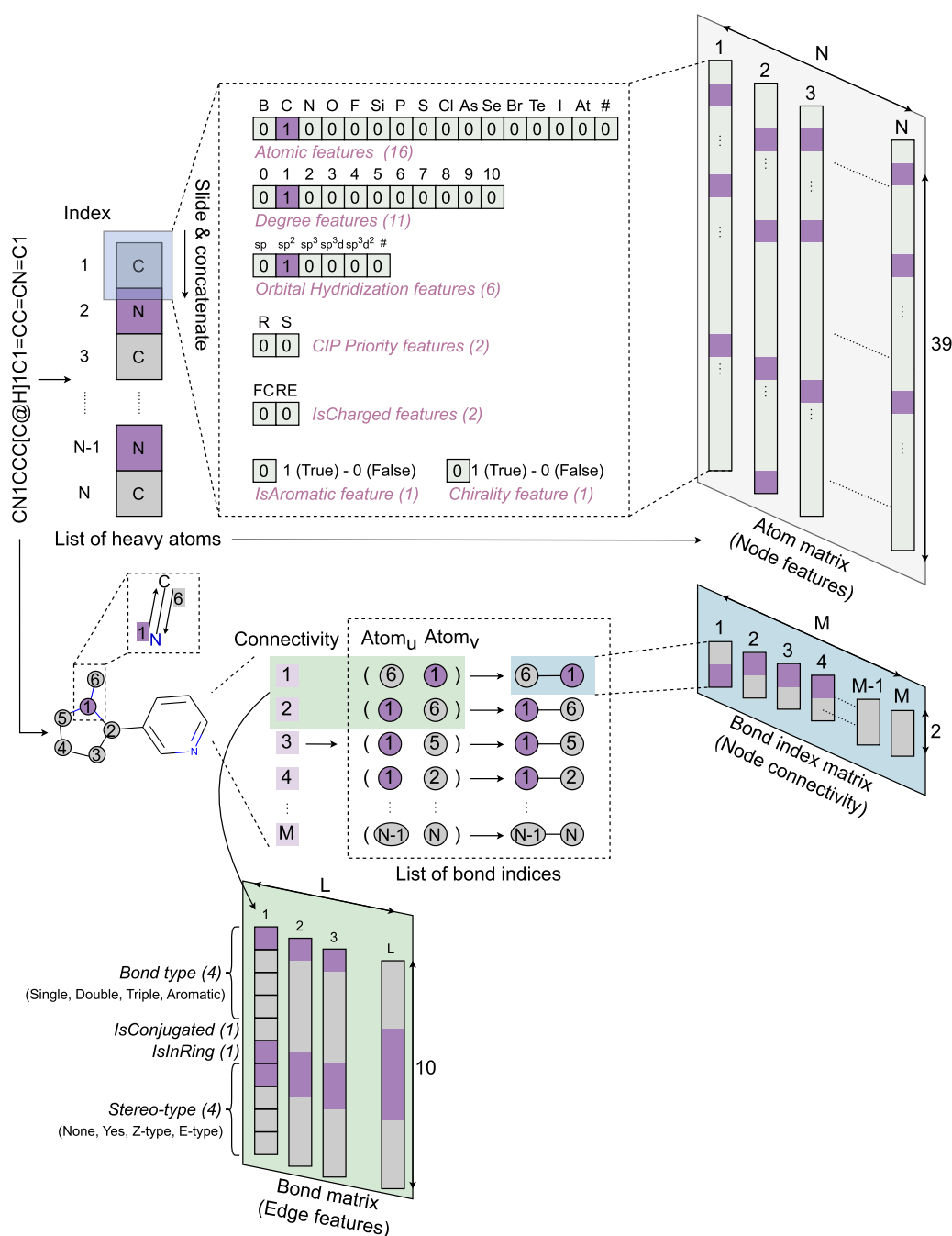


Figure 2. Molecular encoding scheme. An N -atom molecule with M bonds is transformed into an atom matrix of size $N \times 39$, a bond index matrix of size $M \times 2$, and a bond matrix of size $L \times 10$. The atom matrix is a column-wise combination of N vectors of size 39×1 derived from 16 Atomic features, 11 Degree features, 2 IsCharge features, 6 Orbital Hybridization features, 1 IsAromatic feature, 2 Cahn-Ingold-Prelog (CIP) Priority features, and 1 Chirality feature. The bond index matrix is a combination of M vectors of size 2×1 indicating connectivity between atom i th and atom j th. The bond matrix is a combination of L vectors of size 10×1 between atom i th and atom j th.

interconnections of these atoms, and L bonds (two interconnections create one bond), is represented by three matrices: an atom matrix of size $N \times 39$, a bond index matrix of size $M \times 2$, and a bond matrix of size $L \times 10$. It is important to note that the values of N , M , and L may vary across distinct molecules.

The atomic matrix is constructed through a series of steps. Initially, the molecular structure is examined to determine the sequence of heavy atoms within its backbone. Subsequently, these heavy atoms are indexed. For each atom, a set of 39 features is computed by the RDKit library,³⁸ encompassing 16

Atomic features, 11 Degree features, 2 IsCharge features, 6 Orbital Hybridization features, 1 IsAromatic feature, 2 Cahn-Ingold-Prelog (CIP) Priority features, and 1 Chirality feature. These features are merged into a 1×39 vector. The Atomic features identify the atom as B, C, N, O, F, Si, P, S, Cl, As, Se, Br, Te, I, or At or another element. The Degree features represent the number of bonds that the atom forms with adjacent heavy atoms, ranging from 0 to 10. The IsCharge features encode the formal charge state and the radical electron state of the atom. The Orbital Hybridization features denote the type of orbital hybridization employed by the heavy atom to establish its bond.

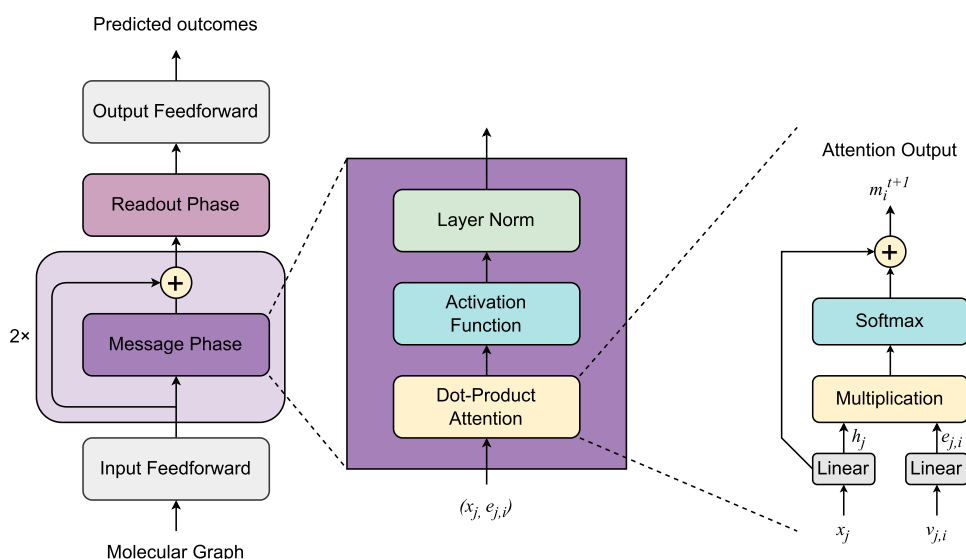


Figure 3. Model architecture of the proposed method.

The IsAromatic features determine whether the atom belongs to any potential ring or cyclic structures. The CIP features indicate the atom's spatial arrangement orientation, either clockwise or counterclockwise when it constitutes a chiral center. The Chirality features determine whether a heavy atom is a chiral center by examining its neighboring connected atoms. The bond index matrix of the molecule is calculated by interpreting the molecular backbone to describe the connectivity map of all heavy atoms (Figure 2). Besides the bond index matrix, the bond matrix is created to determine the type of bonds, conjugation, in-ring position, and type of stereobonds. In our graph-based DL model, the atom matrix, bond matrix, and bond index matrix correspond to the node features, edge features, and node connectivity, respectively.

2.3. Model Architecture. 2.3.1. Message-Passing Phase.

Gilmer et al. surveyed various message computation functions and pointed out that the edge network demonstrated superior performance compared to the others.³⁹ However, this message computation function is characterized by a substantial number of parameters, causing a high computational demand, hence limiting its generalizability across wider contexts. To reduce the computational cost, Li et al. proposed a graph neural network in combination with multihead triplet attention.⁴⁰ Their network demonstrated competitive performance compared to that of the edge network, while the training speed was significantly faster. Li et al.'s network can work more effectively on large data sets by adjusting the number of heads, but it trades off the training speed against the prediction efficiency. A more complex architecture with an increased number of attention heads may be prone to overfitting for small- and medium-sized data sets. To address this issue, we propose a graph neural network specified by the dot-product attention mechanism, which is an efficient attention mechanism.

For the node features $h^t = \{h_0^t, h_1^t, \dots, h_N^t\}$ and edge features e^t at t time step, our network initially maps the source node feature h_j^t and edge features $e_{j,i}^t$ to the same dimension $\mathcal{D}$ before performing a dot-product to obtain a score $S_{j,i}^{t+1}$ at the next time step ($t + 1$) according to

$$S_{j,i}^{t+1} = (W_e \cdot e_{j,i}^t)^T \odot (W_h \cdot h_j^t) \quad (1)$$

where $W_h \in \mathbb{R}^{F1 \times \mathcal{D}}$ and $W_e \in \mathbb{R}^{F2 \times \mathcal{D}}$ are weights shared across all nodes and edges with original dimensions $F1$ and $F2$, respectively. The softmax function is then employed to normalize the coefficients (choices) of neighboring node_{*i*} to obtain the attention score $\alpha_{j,i}^{t+1}$.

$$\alpha_{j,i}^{t+1} = \text{softmax}(S_{j,i}^{t+1}) = \frac{e^{S_{j,i}^{t+1}}}{\sum_{j \in \mathcal{N}_i} e^{S_{j,i}^{t+1}}} \quad (2)$$

where $\mathcal{N}_i$ is the set of neighboring nodes of node_{*i*}. The normalized attention score and source node features h_j are then used to compute the message at the next time step

$$m_i^{t+1} = \sum_{j \in \mathcal{N}_i} \alpha_{j,i}^{t+1} \odot (W_h \cdot h_j^t) \quad (3)$$

2.3.2. Readout Phase. The Set2Set network, introduced by Vinyals et al., was used as the readout function for the generation of graph-level embeddings.⁴¹ This network aggregates node features through distinct attention weights and integrates the resultant features with historical information.

$$q_t = \text{LSTM}(q_t^* - 1) \quad (4)$$

$$\alpha_{i,t} = \text{softmax}(h_i^T \cdot q_t) \quad (5)$$

$$r_t = \sum_{i=1}^N \alpha_{i,t} \cdot h_i^T \quad (6)$$

$$q_t^* = q_t \parallel r_t \quad (7)$$

where $\parallel$ denotes concatenation. The Set2Set network repeatedly executes the computing process for T iterations, resulting in the final graph representation q_t^* . Subsequently, the obtained representation q_t^* is input into a feed-forward neural network to obtain the predicted outcome $\hat{y}$.

2.3.3. Overall Description. Figure 3 illustrates the model architecture of our proposed method. The molecular graph serves as the model's input, which is initially fed into the input feed-forward (IFF) network characterized by a convolutional graph layer. Subsequently, the resulting output matrix is directed through the message-passing phase (MP) network, enabling the

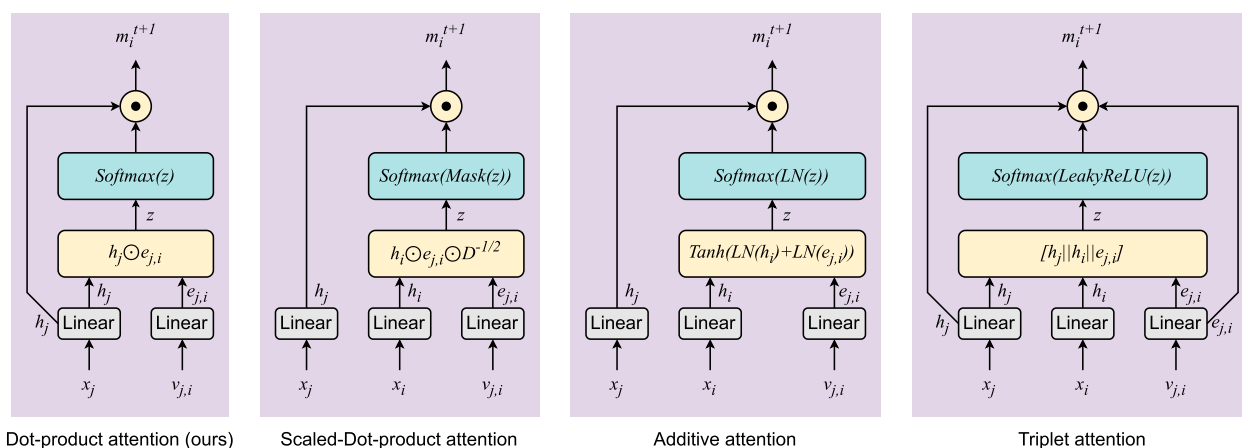


Figure 4. Attention mechanisms used for model benchmarking (LN denotes “linear layer”; $\odot$ denotes matrix multiplication; $\parallel$ denotes concatenation; x_p , x_p , and $v_{j,i}$ are the source node features, target node features, and edge features from node_j to node_p, respectively; h_p , h_p , and $e_{j,i}$ are the source node features, target node features, and edge features from node_j to node_i, respectively, after being linearly normalized).

Table 2. Performance of Models on the Test Set across Four Types of Attention Mechanisms

metric	group	attention mechanism			
		dot-product	scale-dot-product	additive	triplet
AUCROC	cardiac failure	0.7663	0.6746	0.7183	0.7404
	arrhythmia	0.9221	0.8707	0.8468	0.8784
	heart block	0.9379	0.8429	0.8278	0.8485
	myocardial infarction	0.8948	0.8619	0.8679	0.8909
	hypertension	0.9557	0.8794	0.8879	0.8948
	hERG blocker	0.9770	0.9041	0.9020	0.9243
AUCPR	cardiac failure	0.5750	0.4976	0.5085	0.5854
	arrhythmia	0.9104	0.8631	0.8401	0.8566
	heart block	0.9434	0.8563	0.8217	0.8643
	myocardial infarction	0.8764	0.7752	0.8534	0.8821
	hypertension	0.9415	0.8751	0.8824	0.8898
	hERG blocker	0.9790	0.9153	0.9159	0.9360

model to acquire the attention-centered features of the molecules. The outputs of both the MP and IFF networks are combined prior to progressing to the readout phase (RP) network, which is characterized by the Set2Set network.⁴¹ The RP network’s output is passed into the output feed-forward (OFF) networks, ultimately yielding the predicted result. These OFF networks consist of two fully connected layers that are activated by the rectified linear unit function.

2.4. Model Training. Our models were trained using the Adam optimizer⁴² with a learning rate of 1×10^{-4} to iteratively update the network parameters. The binary cross entropy was used as the loss function

$$\text{Loss} = \sum_{i=1}^n y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i) \quad (8)$$

where y_i is the ground truth of data point i th in the n training set and $\hat{y}_i$ is the predicted probability of that data point. Our models were implemented using PyTorch 1.12.0 and trained over 100 epochs using an AMD Ryzen 7-5800X CPU equipped with 32GB RAM and one NVIDIA GeForce RTX 3060 GPU. Model selection was determined by the epoch associated with minimal validation loss. The training duration per epoch ranged from approximately 0.25 to 1.3 s, whereas the testing duration for each model was about 0.01 s. The results of fivefold cross-validation

for all data sets are also provided in Table S1 (Supporting Information).

2.5. Model Evaluation. All the models were evaluated using five common evaluation metrics, including balanced accuracy (BA), F1 score (F1), Matthews correlation coefficient (MCC), the area under the receiver operating characteristic curve (AUCROC), and the area under the precision-recall curve (AUCPR). We conducted three comparative analyses to assess the efficiency of our models against other models developed by using different computational methods or settings.

- The first comparative analysis was to investigate the effectiveness of the MP proposed in our model architecture compared to three other MPs (featured by different attention mechanisms). The model architectures were the same as each other, except for the attention layers. Details of the analysis are provided in Subsection 3.1.
- The second comparative analysis was to benchmark our models against a number of ML models. For each group of prediction tasks, we created 12 ML models using pairwise combinations of four different learning algorithms and three different molecular representations. These ML models were then developed using the same training sets and evaluated on the same test sets as our model. Details of the analysis are provided in Subsection 3.2.

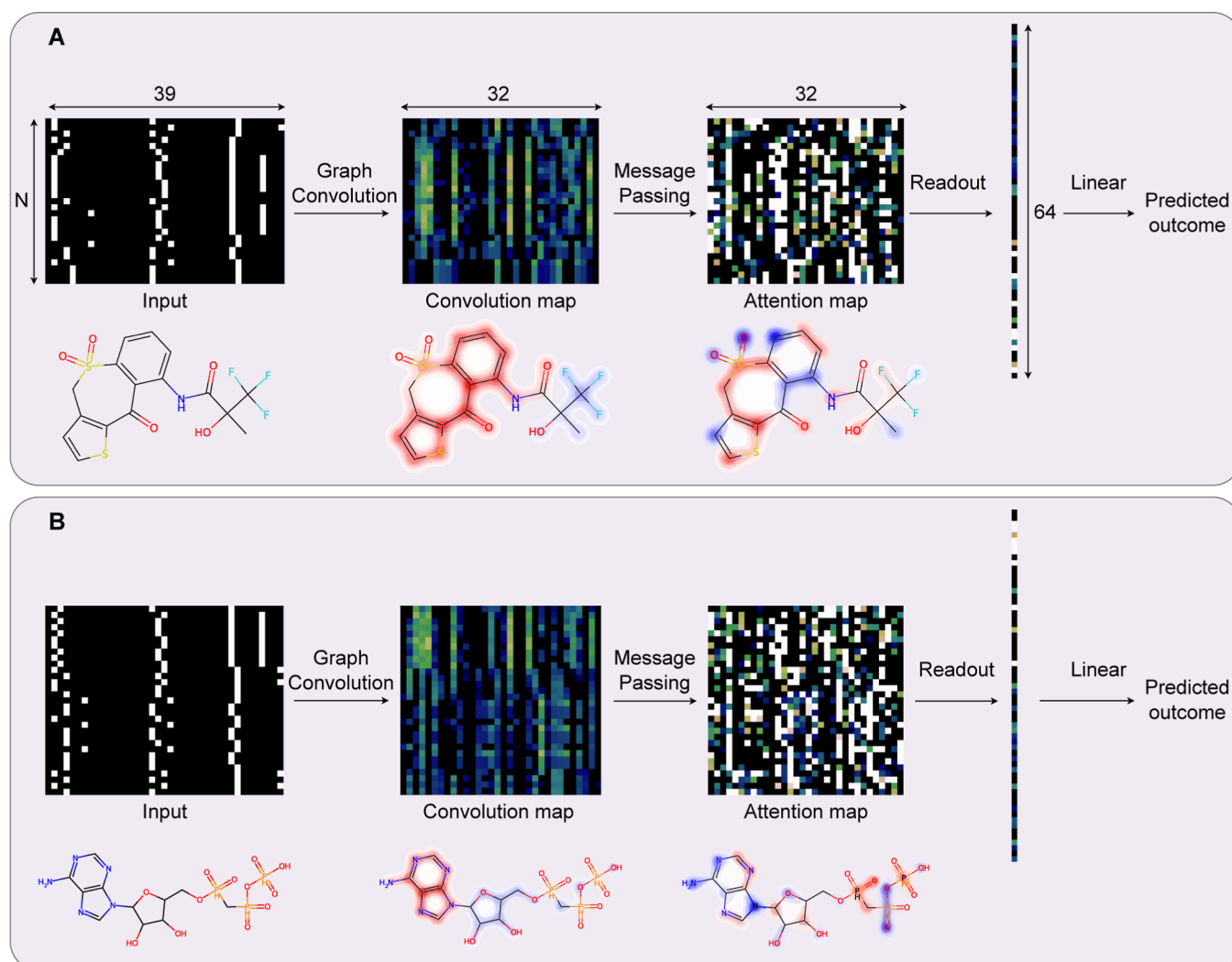


Figure 5. Attention maps learned by the model (A). Negative sample (B). Positive sample. The magnitude of the color indicates the degree of focus on particular substructures. Red-colored and blue-colored regions refer to positive and negative focuses, respectively. Unfocused regions are uncolored.

- The third comparative analysis was to benchmark our models against two advanced graph-based DL models: TrimNet⁴⁰ and graph multiset transformer (GMT).⁴³ For each group of prediction tasks, we created two other DL models using the same graph representation as ours but with different network architectures. These DL models were then developed using the same training sets and evaluated on the same test sets as our models. Details of the analysis are provided in Subsection 3.3.

All of the DL models were trained, optimized, and evaluated under the same conditions, with identical training and test sets used for experiments on each model. The training set was then split into five folds, in which each fold iteratively served as a validation set to train an individual model. The ensemble of the five individual models determines the predictions for each model.

3. RESULTS AND DISCUSSION

3.1. Comparison of Attention Mechanisms. To demonstrate the suitability of the dot-product attention mechanism with our proposed network architecture, we also tested three other attention mechanisms, including the *Scaled-dot-product*,⁴⁴ *Additive*,⁴⁵ and *Triplet*⁴⁰ attentions (Figure 4).

Table 2 summarizes the performance of the models on the test set across the four types of attention mechanisms. The results indicate that our proposed architecture works more effectively when combined with the dot-product attention mechanism, followed by the *Triplet*, *Additive*, and *Scale dot-product* attention mechanisms. In all the attention mechanisms, the source node features x_i were multiplied with the attention scores α to create the attention output m^{t+1} at the time step $(t + 1)$. Compared to the other three mechanisms that use the features of the source node, target node, and edge (x_p, x_t, e_{ji}) , respectively, as MP's inputs, our MP was designed to receive only the features of the source node and edge (x_p, e_{ji}) . The models incorporated with the *dot-product* attention mechanism show higher AUCROC values compared to those of the models using the other mechanisms. The *Triplet* attention mechanism achieved better performance in the “cardiac failure” and “myocardial infarction” groups in terms of AUCPR. Despite being a streamlined attention mechanism, the *dot-product* attention demonstrates itself as the most suitable mechanism that helps the models (under the scope of our study) learn faster while still sufficiently focusing on structural patterns between groups of compounds.

Figure 5 visualizes the convolution and attention maps learned by our model. The node feature matrix is imputed to be

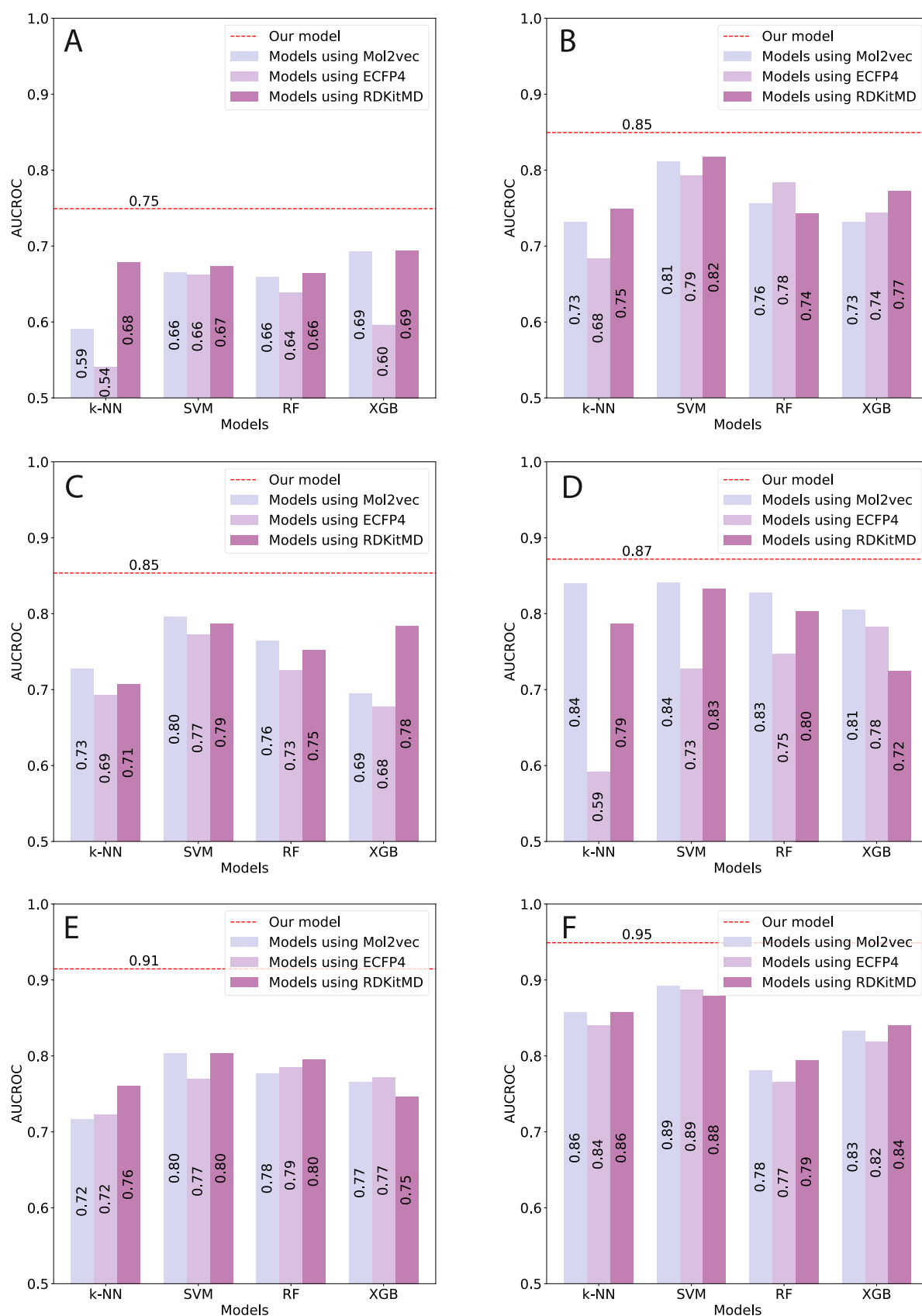


Figure 6. AUCROC values of different models using different molecular presentations: (A) cardiac failure, (B) arrhythmia, (C) heart block, (D) myocardial infarction, (E) hypertension, and (F) hERG blocker.

processed in the graph convolution layer to create a convolution map. The convolution map is then passed through the message-

passing layer to generate an attention map. Finally, the attention map is sent to the readout layer. The output from the readout

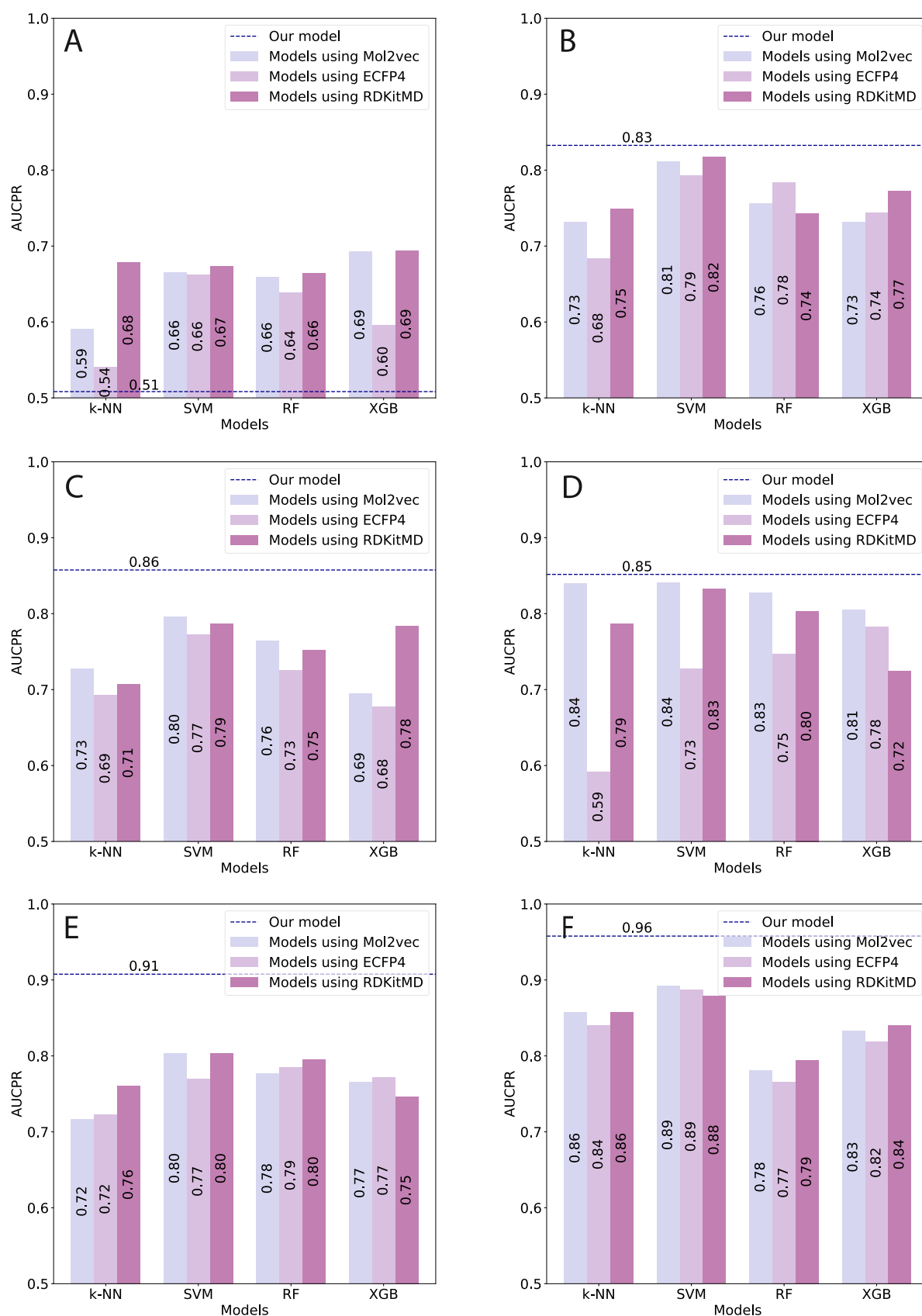


Figure 7. AUCPR values of different models using different molecular presentations: (A) cardiac failure, (B) arrhythmia, (C) heart block, (D) myocardial infarction, (E) hypertension, and (F) hERG blocker.

layer is flattened and activated to return the final prediction. Bright regions are locations at which the model learned to

attend. For each convolution and attention map, the corresponding molecular heatmaps are generated based on the

Table 3. Performances of Our Model and Other Two Graph-Based Deep Learning Models

group	model	metric				
		AUCROC	AUCPR	BA	F1	MCC
cardiac failure	TrimNet	0.6117	0.3363	0.5000	0.0000	0.0000
	GMT	0.7296	0.4805	0.5000	0.0000	0.0000
	CNN	0.6133	0.3310	0.5000	0.0000	0.0000
	DNN	0.5800	0.3160	0.5000	0.0000	0.0000
	ours	0.7492	0.5083	0.6208	0.2739	0.4314
arrhythmia	TrimNet	0.7450	0.7012	0.6628	0.3257	0.6269
	GMT	0.7693	0.7003	0.6984	0.3952	0.6849
	CNN	0.5000	0.7233	0.5000	0.6175	0.0000
	DNN	0.5856	0.5453	0.5530	0.6294	0.1552
	ours	0.8495	0.8326	0.7435	0.4842	0.7273
heart block	TrimNet	0.6808	0.6570	0.6098	0.2632	0.6903
	GTM	0.7702	0.7345	0.7210	0.4674	0.7573
	CNN	0.7253	0.6444	0.5000	0.6617	0.0000
	DNN	0.4939	0.4861	0.5326	0.6613	0.1080
	ours	0.8535	0.8575	0.7434	0.5149	0.7767
myocardial infarction	TrimNet	0.7337	0.6975	0.5897	0.2578	0.6981
	GMT	0.8455	0.8257	0.7436	0.5220	0.7826
	CNN	0.8080	0.7913	0.5000	0.6667	0.0000
	DNN	0.6292	0.6145	0.5641	0.6852	0.2006
	ours	0.8718	0.8517	0.8077	0.6325	0.8276
hypertension	TrimNet	0.7472	0.7602	0.6326	0.3062	0.7108
	GMT	0.8549	0.8345	0.7941	0.5982	0.8138
	CNN	0.7612	0.7692	0.5000	0.6734	0.0000
	DNN	0.5351	0.5462	0.5000	0.6734	0.0000
	ours	0.9146	0.9076	0.8251	0.6529	0.8345
hERG blocker	TrimNet	0.7441	0.7690	0.6684	0.3439	0.7113
	GMT	0.8706	0.8867	0.7744	0.5547	0.7989
	CNN	0.8924	0.8862	0.5000	0.6903	0.0000
	DNN	0.5124	0.5473	0.5000	0.6903	0.0000
	ours	0.9490	0.9578	0.8634	0.7335	0.8784

learned weights. As shown in Figure 5, the convolution layers extract a wide range of possibly important substructures, while the attention layers focus on highly discriminant substructures.

3.2. Comparison with Machine Learning Models. We used the k -nearest neighbors^{46,47} (k -NN),⁴⁸ SVM, random forest⁴⁹, and eXtreme gradient boosting⁵⁰ algorithms in combination with the Mol2Vec⁵¹ features, extended-connectivity fingerprints⁵² (ECFP), and³⁸ RDKitMD to develop 12 different ML models. The grid search method was used to fine-tune the ML models based on fivefold cross-validation. The details of the tuned hyperparameters and their search ranges are provided in Table S2 (Supporting Information). The results indicate that our models on six prediction tasks outperformed the corresponding ML model in terms of AUCROC. Our models are only dominant over other ML models in five prediction tasks (except for the prediction task for the “cardiac failure” group). In the prediction task for the “cardiac failure” group, our models work less effectively compared to other ML models. The small size of the data set and the imbalanced-class issue of this group’s data set may affect the learning of the neural network, especially the attention mechanism. The SVM models exhibit superior efficiency as compared to other ML models since they achieve higher predictive performances on five out of six distinct prediction tasks. Also, the results show that the Mol2vec and RDKitMD features are more adaptive to distance-based learning algorithms, such as k -NN and SVM, compared to tree-based algorithms. There is no large variation among tree-based models trained by binary features (ECFP) and continuous

features. Figures 6 and 7 present the performances of 12 ML models across six groups of molecules. The complete results of all ML models are summarized in Tables S3 and S4 (Supporting Information).

3.3. Comparison with Graph-Based DL Models. We also benchmarked our network architecture against two advanced graph-based deep neural networks (DNNs): TrimNet⁴⁰ and GMT,⁴³ and two baseline neural networks: convolutional neural network (CNN) and DNN. The CNN model was trained using index vectors derived from the ECFP.⁵³ The DNN model was trained using a set of 196 molecular descriptors calculated by RDKit.³⁸ To construct each DL model, including ours, we divided the training set into five stratified folds, and each fold was used as the validation fold, while the remaining folds were used as the training folds. Hence, for each network, we obtained five individual models, and the final predicted probability is derived from the average value of the five predicted probabilities. Applying this ensemble learning strategy can improve the performance of the model, as all of the samples in the training set were used for model training, and the ensemble averages out errors and variance from the individual models, leading to increased stability and robustness. Table 3 summarizes the performances of two graph-based DL models and ours.

Generally, the results indicate that our model works more effectively compared to the TrimNet and GMT models in terms of AUCROC and AUCPR. However, for the prediction task for “cardiac failure”, all three graph-based models work less effectively than ML models due to the small size of the data

set. It should be noted that the TrimNet and GMT models are still highly robust graph-based DNNs that can be applied to both regression and classification problems. However, their complex structure makes them possibly inappropriate for small or medium-sized data sets. In addition, the inclusion of multihead attention in both architectures slows down the training process as the number of heads increases. On the other hand, reducing the number of heads might speed up the training process. However, this decision negatively affects the models' ability to learn and limits the overall potential of the multihead block. Hence, our utilization of a simplified attention mechanism not only expedites the training process but also exhibits higher efficacy in tackling this drug discovery problem compared to more complex techniques.

For the two baseline models, the DNN model works less efficiently than the CNN model. While DNN has the lowest performance in all tasks, the CNN model works more effectively than the TrimNet model in most tasks. Message-passing neural networks (MPNN), such as TrimNet,⁴⁰ AttentiveFP,⁵⁴ and our model, exhibit similar designs characterized by two primary phases: MP and RP. The change in the MP with a modified attention mechanism can drive the learning process differently. TrimNet's contribution is to introduce the triplet message attention mechanism into the MPNN to reduce the number of learnable parameters. In our work, we used MP, which is different from TrimNet.

3.4. Model Stability. To evaluate the stability of the model, we conducted 10 modeling experiments, utilizing distinct pairs of training and test sets in each trial, with the intent of examining the degree of variation and mitigating the impact of random factors. The variation in model performances across the test sets throughout the 10 randomized trials is presented in Table 4. The

Table 4. Performance of Models on the Test Set Over 10 Random Trials

group	metric	
	AUCROC	AUCPR
cardiac failure	0.8333 (0.0628)	0.6560 (0.0998)
arrhythmia	0.8390 (0.0411)	0.7996 (0.0517)
heart block	0.8625 (0.0308)	0.8466 (0.0418)
myocardial infarction	0.8439 (0.0415)	0.8295 (0.0513)
hypertension	0.8631 (0.0397)	0.8477 (0.047)
hERG blocker	0.9469 (0.0069)	0.9531 (0.006)

findings reveal that the average AUCROC values for all prediction tasks surpass 0.8. Although the average AUCPR value associated with the prediction task for the "cardiac failure" group falls below 0.7, the AUCPR values for the remaining tasks span approximately 0.8 to over 0.9. Such outcomes suggest that the performance of the models may be somewhat influenced by the randomness of the data partitioning. Nonetheless, the relatively minor fluctuations in prediction imply that the stability of the models remains within acceptable bounds.

3.5. Limitations and Future Work. While our proposed model demonstrated positive outcomes, it is essential to acknowledge its limitations. Through the experiments, we identified two factors: the imbalance between the positives and negatives and the relatively small size of the data sets that affected the prediction efficiency. These factors have impacts on the overall performance of all of the graph-based models, including ours. To partially address this issue, the utilization of drugs in clinical trials, if available, for external testing could

provide valuable insights for further investigation. Within the scope of this study, we made great efforts to reimplement all of the models. Although the results show that our method outperformed the others, there is a chance that the reimplemented models may not work well with certain input vectors. To fully confirm the effectiveness of a method for molecular property prediction, additional experiments are needed on various data sets with different sizes and tasks (e.g., regression and classification).

4. CONCLUSIONS

In our study, we proposed a graph-based DL model incorporating a simplified attention mechanism to predict the cardiotoxicity of molecules. Our model demonstrated itself as a more suitable approach to address this topic in drug discovery compared to both traditional ML models and advanced graph-based DL models. The incorporation of a simplified attention mechanism expedites the model's training process while maintaining its ability to concentrate on learning specific molecular attributes, thereby optimizing prediction efficacy. The utilization of a graph neural network in conjunction with a simplified attention mechanism may prove to be more effective in comparison to alternative graph-based methods employing complex attention mechanisms, which are predominantly suited for large-sized data sets.

■ ASSOCIATED CONTENT

Data Availability Statement

To support the research community in screening the cardiotoxicity of molecules, we deployed our model as an online web server with a user-friendly interface. The link to the web server and the code with data used in this study are available at <https://github.com/mldlproject/2023-iCardioTox-AGN>.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.3c01286>.

Results from a fivefold cross-validation of the proposed models; search ranges of hyper-parameters used for model tuning; performance of ML models on the independent test sets of "cardiac failure", "arrhythmia", and "heart block" groups; performance of ML models on the independent test sets of "myocardial infarction", "hypertension", and "hERG blocker" groups; chemical space covered by "cardiac failure" samples; chemical space covered by "arrhythmia" samples; chemical space covered by "heart block" samples; chemical space covered by "myocardial infarction" samples; chemical space covered by "hypertension" samples; and chemical space covered by "hERG blocker" samples (PDF)

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Author Contributions

T.V.: data curation, investigation, and writing—original draft. L.N.: data curation, investigation, formal analysis, and validation. Q.H.T.: data curation, investigation, validation, and software. T.-H.N.-V.: conceptualization, methodology, formal analysis, visualization, and writing—review and editing, supervision. B.P.N.: conceptualization, methodology, formal analysis, visualization, and writing—review and editing, supervision. T.V. and L.N. equal contribution.

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Notes

The authors declare no competing financial interest.

REFERENCES

- (1) Smith, P. L.; Baukrowitz, T.; Yellen, G. The inward rectification mechanism of the HERG cardiac potassium channel. *Nature* **1996**, *379*, 833–836.
- (2) Redfern, W.; Carlsson, L.; Davis, A.; Lynch, W.; Mackenzie, I.; Palethorpe, S.; Siegl, P.; Strang, I.; Sullivan, A.; Wallis, R. Relationships between preclinical cardiac electrophysiology, clinical QT interval prolongation and torsade de pointes for a broad range of drugs: evidence for a provisional safety margin in drug development. *Cardiovasc. Res.* **2003**, *58*, 32–45.
- (3) Sanguinetti, M. C.; Jiang, C.; Curran, M. E.; Keating, M. T. A mechanistic link between an inherited and an acquired cardiac arrhythmia: HERG encodes the IKr potassium channel. *Cell* **1995**, *81*, 299–307.
- (4) Brown, A. Drugs, hERG and sudden death. *Cell Calcium* **2004**, *35*, 543–547.
- (5) Aronov, A. Predictive in silico modeling for hERG channel blockers. *Drug Discovery Today* **2005**, *10*, 149–155.
- (6) Villoutreix, B. O.; Taboureau, O. Computational investigations of hERG channel blockers: New insights and current predictive models. *Adv. Drug Delivery Rev.* **2015**, *86*, 72–82.
- (7) Witchel, H. J. The hERG potassium channel as a therapeutic target. *Expert Opin. Ther. Targets* **2007**, *11*, 321–336.
- (8) Raschi, E.; Vasina, V.; Poluzzi, E.; De Ponti, F. The hERG K⁺ channel: target and antitarget strategies in drug development. *Pharmacol. Res.* **2008**, *57*, 181–195.
- (9) Darpo, B.; Nebout, T.; Sager, P. T. Clinical Evaluation of QT/QTc Prolongation and Proarrhythmic Potential for Non-antiarrhythmic Drugs: The International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use E14 Guideline. *J. Clin. Pharmacol.* **2006**, *46*, 498–507.
- (10) Polak, S.; Wiśniowska, B.; Brandys, J. Collation, assessment and analysis of literature in vitro data on hERG receptor blocking potency for subsequent modeling of drugs' cardiotoxic properties. *J. Appl. Toxicol.* **2009**, *29*, 183–206.
- (11) Cheng, F.; Yu, Y.; Shen, J.; Yang, L.; Li, W.; Liu, G.; Lee, P. W.; Tang, Y. Classification of Cytochrome P450 Inhibitors and Non-inhibitors Using Combined Classifiers. *J. Chem. Inf. Model.* **2011**, *51*, 996–1011.
- (12) Cheng, F.; Li, W.; Zhou, Y.; Shen, J.; Wu, Z.; Liu, G.; Lee, P. W.; Tang, Y. admetSAR: A Comprehensive Source and Free Tool for Assessment of Chemical ADMET Properties. *J. Chem. Inf. Model.* **2012**, *52*, 3099–3105.
- (13) Lei, T.; Li, Y.; Song, Y.; Li, D.; Sun, H.; Hou, T. ADMET evaluation in drug discovery: 15. Accurate prediction of rat oral acute toxicity using relevance vector machine and consensus modeling. *J. Cheminf.* **2016**, *8*, 6.
- (14) Lei, T.; Chen, F.; Liu, H.; Sun, H.; Kang, Y.; Li, D.; Li, Y.; Hou, T. ADMET Evaluation in Drug Discovery. Part 17: Development of Quantitative and Qualitative Prediction Models for Chemical-Induced Respiratory Toxicity. *Mol. Pharmaceutics* **2017**, *14*, 2407–2421.
- (15) Lei, T.; Sun, H.; Kang, Y.; Zhu, F.; Liu, H.; Zhou, W.; Wang, Z.; Li, D.; Li, Y.; Hou, T. ADMET Evaluation in Drug Discovery. 18. Reliable Prediction of Chemical-Induced Urinary Tract Toxicity by Boosting Machine Learning Approaches. *Mol. Pharmaceutics* **2017**, *14*, 3935–3953.
- (16) Cai, C.; Fang, J.; Guo, P.; Wang, Q.; Hong, H.; Moslehi, J.; Cheng, F. In Silico Pharmacoeconomic Evaluation of Drug-Induced Cardiovascular Complications Using Combined Classifiers. *J. Chem. Inf. Model.* **2018**, *58*, 943–956.
- (17) Roche, O.; Trube, G.; Zuegge, J.; Pflimlin, P.; Alanine, A.; Schneider, G. A Virtual Screening Method for Prediction of the hERG Potassium Channel Liability of Compound Libraries. *ChemBioChem* **2002**, *3*, 455.
- (18) Bains, W.; Basman, A.; White, C. HERG binding specificity and binding site structure: evidence from a fragment-based evolutionary computing SAR study. *Prog. Biophys. Mol. Biol.* **2004**, *86*, 205–233.
- (19) Sinha, N.; Sen, S. Predicting hERG activities of compounds from their 3D structures: Development and evaluation of a global descriptors based QSAR model. *Eur. J. Med. Chem.* **2011**, *46*, 618–630.
- (20) Broccatelli, F.; Mannhold, R.; Moriconi, A.; Giuli, S.; Carosati, E. QSAR Modeling and Data Mining Link Torsades de Pointes Risk to the Interplay of Extent of Metabolism, Active Transport, and hERG Liability. *Mol. Pharmaceutics* **2012**, *9*, 2290–2301.
- (21) Doddareddy, M. R.; Klaasse, E. C.; Shagufta, I.; Izerman, A. P.; Bender, A. Prospective Validation of a Comprehensive In silico hERG Model and its Applications to Commercial Compound and Drug Databases. *ChemMedChem* **2010**, *5*, 716–729.
- (22) Wang, S.; Sun, H.; Liu, H.; Li, D.; Li, Y.; Hou, T. ADMET Evaluation in Drug Discovery. 16. Predicting hERG Blockers by Combining Multiple Pharmacophores and Machine Learning Approaches. *Mol. Pharmaceutics* **2016**, *13*, 2855–2866.
- (23) Cai, C.; Guo, P.; Zhou, Y.; Zhou, J.; Wang, Q.; Zhang, F.; Fang, J.; Cheng, F. Deep Learning-Based Prediction of Drug-Induced Cardiotoxicity. *J. Chem. Inf. Model.* **2019**, *59*, 1073–1084.
- (24) Pires, D. E. V.; Blundell, T. L.; Ascher, D. B. pkCSM: Predicting Small-Molecule Pharmacokinetic and Toxicity Properties Using Graph-Based Signatures. *J. Med. Chem.* **2015**, *58*, 4066–4072.
- (25) Ryu, J. Y.; Lee, M. Y.; Lee, J. H.; Lee, B. H.; Oh, K.-S. DeepHIT: a deep learning framework for prediction of hERG-induced cardiotoxicity. *Bioinformatics* **2020**, *36*, 3049–3055.
- (26) Karim, A.; Lee, M.; Balle, T.; Sattar, A. CardioTox net: a robust predictor for hERG channel blockade based on deep learning meta-feature ensembles. *J. Cheminf.* **2021**, *13*, 60.
- (27) Iftkhar, S.; de Sá, A. G. C.; Velloso, J. P. L.; Aljarf, R.; Pires, D. E. V.; Ascher, D. B. cardioToxCsM: A Web Server for Predicting Cardiotoxicity of Small Molecules. *J. Chem. Inf. Model.* **2022**, *62*, 4827–4836.
- (28) Inamdar, A.; Inamdar, A. Heart Failure: Diagnosis, Management and Utilization. *J. Clin. Med.* **2016**, *5*, 62.

- (29) Antzelevitch, C.; Burashnikov, A. Overview of Basic Mechanisms of Cardiac Arrhythmia. *Card. Electrophysiol. Clin.* **2011**, 3, 23–45.
- (30) Yang, Y. C.; Pata, R. K.; Aung, T. T. A Case of Complete Heart Block With Diagnostic Challenge and Therapeutic Dilemma. *J. Invest. Med. High Impact Case Rep.* **2018**, 6, 232470961878811.
- (31) Saleh, M.; Ambrose, J. A. Understanding myocardial infarction. *F1000Research* **2018**, 7, 1378.
- (32) Garbers, D. L.; Dubois, S. K. The Molecular Basis of Hypertension. *Annu. Rev. Biochem.* **1999**, 68, 127–155.
- (33) Garrido, A.; Lepailleur, A.; Mignani, S. M.; Dallemagne, P.; Rochais, C. hERG toxicity assessment: Useful guidelines for drug design. *Eur. J. Med. Chem.* **2020**, 195, 112290.
- (34) Munawar, S.; Windley, M. J.; Tse, E. G.; Todd, M. H.; Hill, A. P.; Vandenberg, J. I.; Jabeen, I. Experimentally Validated Pharmacoinformatics Approach to Predict hERG Inhibition Potential of New Chemical Entities. *Front. Pharmacol.* **2018**, 9, 1035.
- (35) Lipinski, C. A. *Annual Reports in Computational Chemistry*; Elsevier, 2005; pp 155–168.
- (36) Fourches, D.; Muratov, E.; Tropsha, A. Trust, But Verify: On the Importance of Chemical Structure Curation in Cheminformatics and QSAR Modeling Research. *J. Chem. Inf. Model.* **2010**, 50, 1189–1204.
- (37) Nguyen, L.; Nguyen Vo, T.-H.; Trinh, Q. H.; Nguyen, B. H.; Nguyen-Hoang, P.-U.; Le, L.; Nguyen, B. P. iANP-EC: Identifying Anticancer Natural Products Using Ensemble Learning Incorporated with Evolutionary Computation. *J. Chem. Inf. Model.* **2022**, 62, 5080–5089.
- (38) Landrum, G. A.; Tosco, P.; Kelley, B.; Sriniker, Ric; Geddeck; Vianello, R.; NadineSchneider; Dalke, A.; Kawashima, E.; Dan, N.; Cole, B. J.; Swain, M.; Turk, S.; Cosgrove, D. O.; Savelyev, A.; Vaucher, A. C.; Jones, G.; Wójcikowski, M.; Probst, D.; Guillaume Godin; Scalfani, V. F.; Pahl, A.; Berenger, F.; Jlvargo; Strets; Gavid, D.; Sfora, G.; Jensen, J. H. *RDKit: Open-Source Cheminformatics Software (Release 2022.03.2)*, 2022. <http://www.rdkit.org>. Accessed on 06.04.2023.
- (39) Gilmer, J.; Schoenholz, S. S.; Riley, P. F.; Vinyals, O.; Dahl, G. E. *Neural Message Passing for Quantum Chemistry*. In *ICML'17: International Conference on Machine Learning*, 2017; pp 1263–1272.
- (40) Li, P.; Li, Y.; Hsieh, C.-Y.; Zhang, S.; Liu, X.; Liu, H.; Song, S.; Yao, X. TrimNet: learning molecular representation from triplet messages for biomedicine. *Briefings Bioinf.* **2021**, 22, bbaa266.
- (41) Vinyals, O.; Bengio, S.; Kudlur, M. Order Matters: Sequence to Sequence for Sets. *arXiv* **2015**, arXiv:1511.06391. <https://arxiv.org/abs/1511.06391>
- (42) Kingma, D. P.; Ba, J. Adam: A Method for Stochastic Optimization. *arXiv* **2014**, arXiv:1412.6980.
- (43) Baek, J.; Kang, M.; Hwang, S. J. Accurate learning of graph representations with graph multiset pooling. *arXiv* **2021**, arXiv:2102.11533. <https://arxiv.org/abs/2102.11533>
- (44) Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; Polosukhin, I. Attention Is All You Need. *arXiv* **2017**, arXiv:1706.03762. <https://arxiv.org/abs/1706.03762>
- (45) Bahdanau, D.; Cho, K.; Bengio, Y. Neural Machine Translation by Jointly Learning to Align and Translate. *arXiv* **2014**, arXiv:1409.0473. <https://arxiv.org/abs/1409.0473>
- (46) Fix, E.; Hodges, J. L. Discriminatory Analysis. Nonparametric Discrimination: Consistency Properties. *Int. Stat. Rev.* **1989**, 57, 238.
- (47) Altman, N. S. An Introduction to Kernel and Nearest-Neighbor Nonparametric Regression. *Am. Stat.* **1992**, 46, 175.
- (48) Cortes, C.; Vapnik, V. Support-vector networks. *Mach. Learn.* **1995**, 20, 273–297.
- (49) Breiman, L. Random Forests. *Mach. Learn.* **2001**, 45, 5–32.
- (50) Chen, T.; Guestrin, C. XGBoost. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, **2016**.
- (51) Jaeger, S.; Fulle, S.; Turk, S. Mol2vec: Unsupervised Machine Learning Approach with Chemical Intuition. *J. Chem. Inf. Model.* **2018**, 58, 27–35.
- (52) Rogers, D.; Hahn, M. Extended-Connectivity Fingerprints. *J. Chem. Inf. Model.* **2010**, 50, 742–754.
- (53) Nguyen-Vo, T.-H.; Trinh, Q. H.; Nguyen, L.; Nguyen-Hoang, P.-U.; Nguyen, T.-N.; Nguyen, D. T.; Nguyen, B. P.; Le, L. iCYP-MFE: Identifying Human Cytochrome P450 Inhibitors Using Multitask Learning and Molecular Fingerprint-Embedded Encoding. *J. Chem. Inf. Model.* **2022**, 62, 5059–5068.
- (54) Xiong, Z.; Wang, D.; Liu, X.; Zhong, F.; Wan, X.; Li, X.; Li, Z.; Luo, X.; Chen, K.; Jiang, H.; Zheng, M. Pushing the Boundaries of Molecular Representation for Drug Discovery with the Graph Attention Mechanism. *J. Med. Chem.* **2020**, 63, 8749–8760.



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